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Digression lecture 1:

Patterns are everywhere in math. When you boil math down to its essence, all that remains are structures and the patterns they produce.

learning to notice these patterns can lend great insight and intuition into the problem solving process.

This lecture will touch on a few of my favorite 'Math Memes', or patterns that recur often.

Meme #1: algabra tricks

Try manipulating the equation by either $\boxed{\text{adding } 0}$ or $\boxed{\text{multiplying by } 1}$

adding 0 example: recall $a+0=a$ for all a

try to graph: $x^2+10x+16$ without a calculator

... not easily factorable ...

But: $x^2+10x+25=(x+5)^2$ which is only 9 away from the original equation!

so add 9, then subtract 9:

$x^2+10x+\underbrace{16+9}-9$

$=(x^2+10x+25)-9$

$=(x+5)^2-9$ which is much easier to graph, and is equal to the original equation!

Multiplying By 1: recall $a \cdot 1 = a$ forall a.

try multiplying By a convenient choice of 1!

Example: adding fractions

$$\frac{5}{7} + \frac{2}{3} = ?$$

we all know how to solve this, But note were just multiplying By 1!

$$\frac{5}{7} \cdot \underbrace{\frac{3}{3}}_{=1} + \frac{2}{3} \cdot \frac{7}{7} = \frac{15}{21} + \frac{14}{21} = \frac{29}{21} \checkmark$$

$e^{\ln(x)}$: recall forall x (even complex x)

$$e^{\ln(x)} = x.$$

Example: $5^{2x+1} = 100$

Cant do much with this on inspection.

try: $e^{\ln(5^{2x+1})} = e^{(2x+1)\ln(5)} = 100$

then take $\ln$: $(2x+1) \cdot \ln 5 = \ln(100)$

$$\Rightarrow 2x+1 = \frac{\ln(100)}{\ln(5)}$$

$$\Rightarrow x = \frac{1}{2}\left(\frac{\ln(100)}{\ln(5)} - 1\right)$$

the key idea is if you find an equation you can't immediately solve, there are a couple of simple tricks to make it more friendly.